

DBMS Assignment 1

Design and Implementation of a database management system for Research Project, Publication and Grant Management System

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Problem Statement

Research-related information is often maintained across separate files and systems.

- No single view of overall project progress
- Grant and funding details may be scattered across records.
- Project milestones and expenditure are difficult to track.
- Manual data management can lead to duplication, inconsistency and missing information.

Proposed solution to the problem statement.

A centralized relational database system that connects researchers, projects, funding, milestones, expenditure and research outputs.

Project Objectives

- Centralize research project-related information.
- Maintain researcher and department records.
- Manage projects and investigator assignments.
- Track grants, funding agencies and budgets.
- Monitor project milestones and expenditure.
- Link projects with publications and patents.
- Generate meaningful research and funding reports.
- Maintain data integrity and consistency through relational database design.

Scope of the Project

- Register and maintain researcher, department and project details.
- Assign investigators to research projects.
- Manage funding agencies, grants and project budgets.
- Track project milestones and completion status.
- Record and monitor project expenditure and budget utilization.
- Manage research publications, patents and reports.
- Retrieve research information and generate project, funding and research-output reports.

System Users

1. Researcher

- View assigned projects
- Track project milestones
- Edit research outputs & Access publication information

2. Administrator

- Register projects and researchers
- Assign investigators
- Manage grants and funding
- Record milestones & Generate reports

3. Department

- Monitor, Track and Review the research outputs

Functional Requirements

The system should allow authorized users to:

- Register researchers and departments
- Create and manage research projects
- Assign investigators to projects
- Register funding agencies
- Record grants and project budgets
- Define and track milestones
- Record project expenditure
- Link publications to projects
- Maintain patent information
- Generate research and funding reports



Thank You